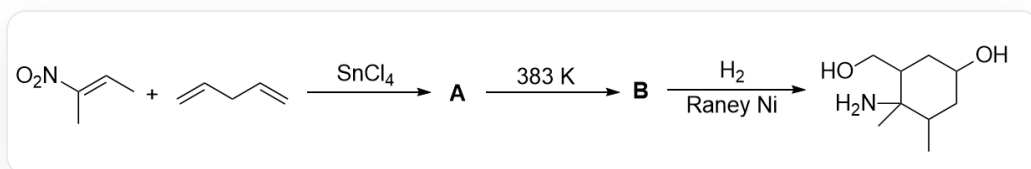


Question



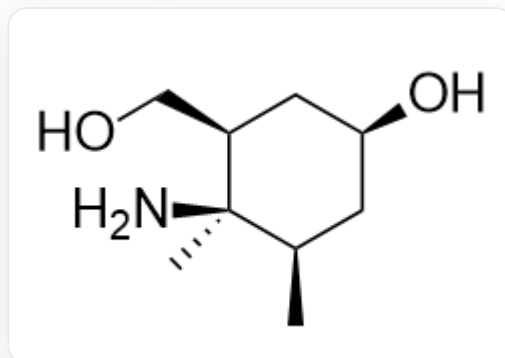
The image describes a one-pot organic cascade reaction. The substrates are C/C([N+])([O-])=O=C\C and C=CCC=C, reacting under the action of SnCl4 to obtain **A**; **A** yields **B** at 383 K; **B** yields the final product OC1CC(C)C(N)(C)C(CO)C1 under H2, RaneyNi conditions.

In the reaction above, it is known that the process of generating the final product from **B** does not involve changes in carbon-carbon bond formation. **A** is the thermodynamically more stable product.

Which of the following options reflects the correct stereochemistry of the final product?

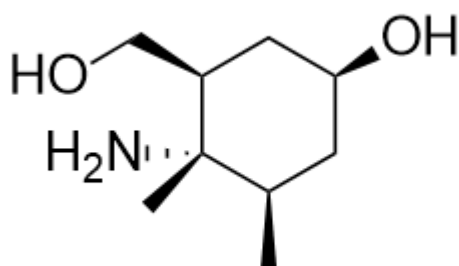
A. All other options are incorrect

B.



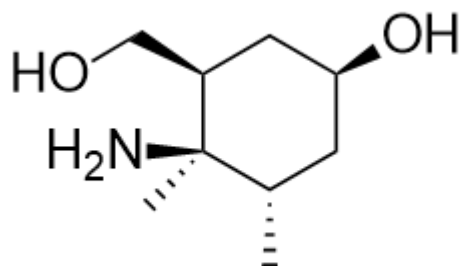
O[C@H]1C[C@@H](C)[C@](N)(C)[C@@H](CO)C1

C.



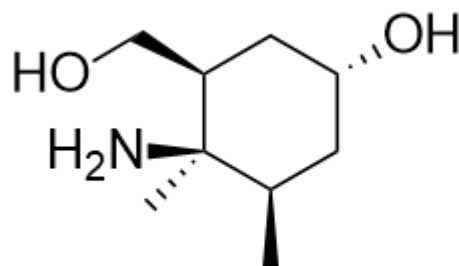
O[C@H]1C[C@@H](C)[C@@](N)(C)[C@@H](CO)C1

D.



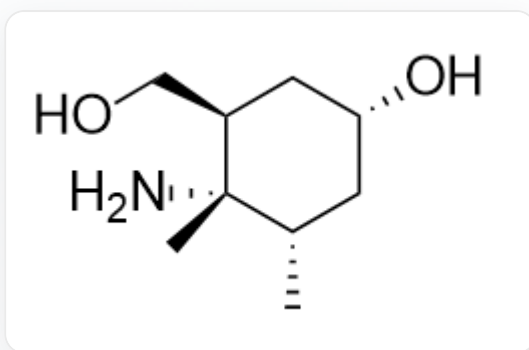
O[C@H]1C[C@H](C)[C@](N)(C)[C@@H](CO)C1

E.



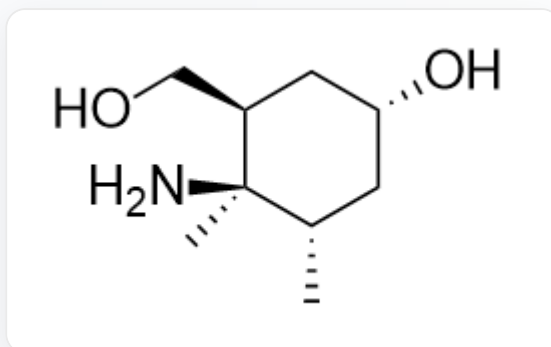
O[C@@H]1C[C@@H](C)[C@](N)(C)[C@@H](CO)C1

F.



O[C@@H]1C[C@H](C)[C@@](N)(C)[C@@H](CO)C1

G.

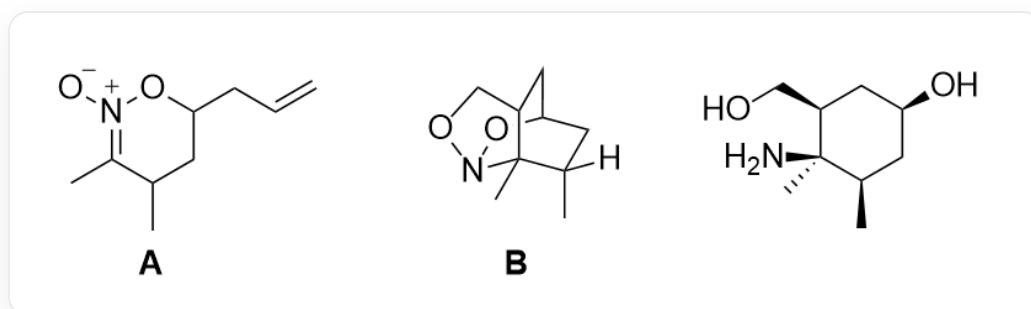


O[C@@H]1C[C@H](C)[C@](N)(C)[C@@H](CO)C1

Answer

Correct Answer: **B**

Detailed Explanation



A structure is [O-][N+]1=C(C)C(C)CC(CC=C)O1; the structure of **B** is CC1(C(CO2)C3)N2OC3CC1([H])C; the product structure is O[C@H]1C[C@@H](C)[C@](N)(C)[C@@H](CO)C1.

The substrate is obviously a dienophile and a diene, so it undergoes a D-A cycloaddition reaction under the catalysis of Lewis acid SnCl4, so the structure of **A** is [O-][N+]1=C(C)C(C)CC(CC=C)O1.

CHECKPOINT

1 PTS

The substrate undergoes a D-A cycloaddition reaction under the catalysis of Lewis acid SnCl4

CHECKPOINT

1 PTS

The structure of **A** is [O-][N+]1=C(C)C(C)CC(CC=C)O1

A is heated to 383 K and converted to **B**. The process of generating the final product from **B** does not involve changes in carbon-atom bonding, so it can only be the breaking of the N – O bond; therefore, without considering stereochemistry, the structure of **B** is such that the nitrogen atom of the final product is connected to two hydroxyl groups, forming a bridged ring structure with the structural formula CC1(C(CO2)C3)N2OC3CC1([H])C.

CHECKPOINT

1 PTS

The process of generating the final product from **B** is the breaking of the N – O bond

CHECKPOINT

1 PTS

The structure of **B** is such that the nitrogen atom of the final product is connected to two hydroxyl groups, forming a bridged ring structure

CHECKPOINT

1 PTS

The structural formula of **B** is CC1(C(CO2)C3)N2OC3CC1([H])C

From this, it can be judged that the reaction of **A** heating to 383 K to convert to **B** is an intramolecular [3+2] cycloaddition reaction; the alkene in **A** undergoes [3+2] cycloaddition with the $C = N^+ - O^-$ structure.

CHECKPOINT

1 PTS

The reaction of **A** converting to **B** is an intramolecular [3+2] cycloaddition reaction

Considering stereochemistry, after the [3+2] cycloaddition, when the bridged ring structure breaks the N – O bond, the generated hydroxymethyl and hydroxyl group were originally on the same side of the bridged ring, so they are in the cis configuration in the final product;

CHECKPOINT

2 PTS

Hydroxymethyl and hydroxyl are in the cis configuration

A is the thermodynamic product. Due to steric hindrance effects, the allyl and methyl groups in **A** should be in the trans conformation. After the [3+2] cycloaddition, the two methyl groups are therefore in the trans conformation; in the final product, they are also in the trans conformation;

CHECKPOINT

1 PTS

In the thermodynamic product **A**, the allyl and methyl groups should be in the trans conformation

CHECKPOINT

2 PTS

The two methyl groups are in the trans conformation

The amino group and the hydroxymethyl group jointly form a five-membered ring of the bridged ring, located on the same side of the cyclohexane ring, so they should be in cis form in the final product.

CHECKPOINT

2 PTS

The amino group and hydroxymethyl group are in the cis configuration

The product structure that satisfies these three stereochemical requirements is O[C@H]1C[C@@H](C)[C@](N)(C)[C@@H](CO)C1, so option B is correct.

CHECKPOINT

1 PTS

The product structure is O[C@H]1C[C@@H](C)[C@](N)(C)[C@@H](CO)C1